

Detail-oriented, organized, and meticulous employee. Works at fast pace to meet tight deadlines. Enthusiastic team player ready to contribute to company success.

## Work History

### Senior Software Engineer *Capital One, Lakeland, FL | January 2021 - November 2024*

- Architected and developed the backend using **Python** and **Django**, and handled frontend development with **React**, **Webpack**, **Redux**, **ES6/7**, and **GraphQL** for efficient database interaction.
- Led the analysis, design, and development of **market risk applications** using Python, integrating them effectively with other services.
- Worked on both frontend and backend modules utilizing Python with the **Django** Web Framework to provide a seamless experience across platforms.
- Built and structured the application according to Angular-style guides and best practices, ensuring scalability and maintainability.
- Developed complex database objects such as stored procedures, functions, packages, and triggers using **SQL** and **PL/SQL** to ensure optimal data management and manipulation.
- Worked on backend development using web services with **Python** (**Flask**, **Django**, **SQLAlchemy**), implementing **RESTful APIs** and ensuring efficient data handling.
- Created APIs, database models, and views utilizing Python to build responsive web applications and improve user interaction with the platform.
- Leveraged **Django ORM** module to write complex database queries, improving data retrieval performance and reducing the load on the database.
- Developed **Python** scripts to parse **XML** documents and load data into databases, optimizing data processing workflows.
- Migrated production infrastructure to Amazon Web Services (**AWS**), leveraging **AWS CloudFormation**, **CodeDeploy**, **EBS**, and **OpsWorks** to automate and streamline deployment and scaling.
- Optimized **ETL** processes by implementing load balancing strategies, database performance tuning, and utilizing advanced ETL processing tools to improve data throughput and reduce latency.
- Implemented automation tools including **Chef**, **Puppet**, and **Splunk** to streamline deployment processes and improve monitoring and log management in cloud environments.
- Developed **Python** scripts to automate database content updates and file manipulations, streamlining workflow and improving operational efficiency.
- Applied object-oriented design principles in **Python** for building maintainable code, improving code quality through proper logging, monitoring, and debugging to ensure optimized performance.
- Designed and developed views and templates using **Django's** view controller and template language to create intuitive and user-friendly website interfaces.
- Created dynamic user interface components using **AngularJS** and **ReactJS**, enhancing the user experience.

- Developed front-end components using **React.js**, including **JSX**, creating React components, working with the **Virtual DOM**, and utilizing React's props, lifecycle methods, states, and event handling.
- Implemented client-side interactivity using **jQuery** for dynamic **DOM** manipulation, enhancing user experience on web applications.
- Developed multiple **React-Redux**-based pages with functionality for login/logout, display of summary metrics, and dynamic content updates.
- Implemented user interface guidelines and standards throughout the development and maintenance process using **HTML**, **CSS**, **JavaScript**, and **jQuery** to create a consistent user experience.
- Performed data cleaning and feature selection using the **MLlib** package in **PySpark** to improve the quality and performance of data-driven applications.
- Conducted data cleaning and ensured data quality, consistency, and integrity using tools like Pandas and **NumPy**, preparing datasets for further analysis and reporting.
- Utilized Python libraries such as **NumPy**, **Matplotlib**, **Pickle**, **Pandas**, **SciPy**, **wxPython**, **PyTables**, and **PyQt** for **data analysis**, **visualization**, and creating complex graphical reports, including histograms and data plots.
- Created a **Lambda** deployment function and configured it to automatically receive events from an **S3 bucket**, streamlining serverless processing.
- Automated deployment processes using **Ansible** and managed application deployment in an **AWS** environment to improve efficiency and reduce errors.
- Managed application environments using **Ansible**, **Vagrant**, and **Docker**, ensuring consistent and scalable development, testing, and production environments.
- Migrated production infrastructure to Amazon Web Services (**AWS**), leveraging **AWS CloudFormation**, **CodeDeploy**, **EBS**, and **OpsWorks** to automate and streamline deployment and scaling.
- Managed source code with **GitHub**, automated build processes using **Jenkins**, and deployed applications in **Docker** containers on **Mesos** to streamline continuous integration and delivery.
- Performed unit testing and integrated testing using frameworks like Python's **unittest** and **pytest** to ensure reliable, bug-free code in production.
- Optimized **Django APIs** through profiling techniques, improving the response time and ensuring faster rendering of information to the frontend.
- Identified and resolved issues in applications, continually improving processes to ensure the delivery of stable, accurate, and efficient solutions.
- **Led a team of 6 members** in the development of a marketplace application, overseeing both backend and frontend development.
- Responsible for software architecture and full-stack development, involving both **backend** and **frontend** technologies, as well as contributing to testing and documentation processes.
- Collaborated with data engineers and operations teams to implement **ETL** processes, writing and optimizing SQL queries for efficient data extraction that met analytical requirements.
- Collaborated with internal and external teams to address issues, provide feedback, and ensure successful project completion.

#### Full Stack Engineer *Capco, Orlando, FL | January 2018- November 2020*

- Led and mentored a team of 7 members, including 2 backend programmers, 1 quality analyst, 2 testers, 1 documentation specialist, 1

team leader, and 1 DevOps engineer, while performing backend development, database design, and training responsibilities.

- Designed and developed web-based applications using **Python**, **Django**, and **PySpark** for big data analysis, focusing on performance optimization and scalability.
- Developed Python modules to connect with and monitor Apache Cassandra instances, improving data access and reliability.
- Developed Python scripts to parse and process **XML documents**, ensuring smooth data migration to databases for efficient storage and access.
- Designed and implemented APIs, database models, and views to build a responsive web application, ensuring a seamless user experience.
- Utilized Python libraries like **NumPy** and **Matplotlib** for data analysis and visualization, providing actionable insights for stakeholders.
- Utilized **Python** and Django for developing graphics, **XML** processing, data exchange, and implementing business logic to streamline operations and enhance user experience.
- Designed and developed database models and views using Python, creating interactive web solutions to streamline data access and user interactions.
- Developed a **Python-based Django web application**, utilizing **Python** scripting for data processing and **MySQL** for database management.
- Developed unit and regression testing frameworks to ensure code reliability and functionality.
- Used **PyTest** to write comprehensive test cases, validating the integrity and quality of newly developed features and bug fixes.
- Wrote and optimized **SQL** queries, stored procedures, and triggers using **Oracle**, **SQL**, and **PL/SQL** for efficient database management.
- Ensured robust data handling and processing by implementing well-designed database operations.
- Managed large datasets using pandas **DataFrames** and **MySQL**, ensuring efficient database operations and data retrieval.
- Converted ETL operations to Hadoop systems using Hive transformations and functions, optimizing data processing pipelines.
- Worked with **AWS services** (S3, EC2, Lambda, RDS, DynamoDB) to implement cloud solutions and automate workflows, ensuring high availability and cost efficiency.
- Configured and deployed web applications on AWS Cloud, creating EC2 instances for scalable infrastructure.
- Deployed Django applications using **Amazon Elastic Beanstalk** and **EC2** instances for scalable application hosting.
- Supported deployment workflows and continuous integration using **GitLab**, **GitLab CI**, **Rundeck**, **Docker**, and **SaltStack** provisioning, enhancing the development cycle and automation.
- Implemented continuous integration with Jenkins for automated deployment on **AWS** infrastructure.
- Conducted systems design and feasibility studies, recommending cost-effective cloud solutions with Amazon Web Services (AWS) to meet business needs and ensure scalability.
- Performed quantitative analysis and data mining to extract valuable insights, trends, and patterns, improving data-driven decision-making.
- Analyzed Hadoop clusters and worked with Big Data components, including **Pig**, **Hive**, **Spark**, **HBase**, **Kafka**, and Elasticsearch.
- Managed data processing tasks within the Hadoop ecosystem, optimizing workflows for large-scale data analysis.
- Developed and optimized data processing tasks using **PySpark**, including reading, merging, enriching, and loading data from external sources into target destinations.

- Worked with large datasets to perform efficient data transformations and preparation for machine learning applications.
- Developed and executed **Python-based** streaming jobs to process terabytes of real-time data for machine learning applications.
- Leveraged data pipelines to enhance the scalability and performance of data processing systems.
- Implemented data wrangling techniques to prepare datasets for advanced analytics and machine learning models.
- Developed front-end web applications using HTML/CSS, jQuery, and Bootstrap, and back-end development with **Golang** and **SQL** to create dynamic, full-stack solutions.
- Designed and developed Django forms for recording user data, enabling real-time data collection and processing on web applications.
- Developed a data-driven dashboard using **Python, Bootstrap, CSS**, and JavaScript for both functionality and responsiveness.
- Ensured the dashboard's seamless operation and usability through comprehensive testing.
- Utilized **HTML, CSS**, and **jQuery** for dynamic data visualization and interactive front-end elements.
- Implemented Object-Oriented Programming (**OOP**) concepts in Python, including creating classes, constructors, overloading, and modules to ensure maintainable and scalable code.
- Built **Django** forms for recording user data, enabling real-time data collection and processing on web applications.
- Developed and tested key features for a data-driven dashboard using **Python, Bootstrap, CSS**, and JavaScript for both functionality and responsiveness.
- Ensured high code quality through consistent use of version control tools such as Subversion to streamline team collaboration.
- Used integrated development environments (**IDE**) and collaborated on the project using JIRA for effective bug tracking and issue resolution.
- Developed Python-based scripts for containerizing analysis tools using Docker and pulling images from Docker Hub, improving deployment and scalability.
- Developed Python modules to monitor and connect to **Apache Cassandra**, ensuring the reliability of distributed database operations.
- Developed Python scripts to parse and process **XML** documents, ensuring smooth data migration to databases for efficient storage and access.
- Executed User Acceptance Testing (**UAT**) as part of the test plan to ensure the application met user requirements.
- Conducted systems design and feasibility studies, recommending cost-effective cloud solutions with AWS to meet business needs and ensure scalability.
- Developed unit and regression testing frameworks to ensure code reliability and functionality.

#### Software Developer *IBM, Doral, FL | January 2013- December 2017*

- Developed consumer-oriented features and applications with a focus on Python, Django, and HTML, utilizing **Behavior-Driven Development (BDD)** and pair programming methodologies.
- Contributed to the development of web-based applications, utilizing a wide range of technologies including **Python 2.7, Django 1.4/1.3, JavaScript, jQuery, XML, CSS3, HTML5**, and **DHTML**.
- Designed and developed REST APIs for web applications related to insurance premium calculations, using the Django REST framework.

- Designed and implemented custom **Bash** scripts for managing and controlling data flow between the application and the **PostgreSQL** database.
- Demonstrated strong knowledge of RDBMS, including hands-on experience with **PostgreSQL**, **Web technologies**, **XML**, **HTML**, **Python**, and **Shell Scripting**.
- Engaged in all stages of the development process, from initial product ideation to scaling in a production environment.
- Led software development best practices for the team, providing mentorship and guidance to junior engineers.
- Collaborated with cross-functional teams, including project managers and higher management from **Sales** and **Account Management**, to understand business requirements and craft efficient, elegant solutions for their implementation.
- Responsible for incorporating the best technologies into the existing system and services to continuously improve and add new features.
- Extensively leveraged **Python's graphics** APIs to create custom graphic solutions and serialization libraries for encoding data in **XML** and **JSON** formats.
- Utilized **Django configurations** to manage URLs and application parameters for seamless integration.
- Assisted the development team in correctly sending data via query strings, ensuring efficient data handling in the backend through **PostgreSQL**.
- **Troubleshoot** and resolved issues related to process execution, collaborating with the team to ensure smooth operation and functionality.

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## Education

**Bachelor of Science in Computer Science** University of South Florida, Tampa, FL | May 2012